BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 190 · Programming constructs and scope

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is the approximate growth of a loop executed n times?

Q2. Why must binary search update low=mid+1 rather than low=mid in the greater-than case?

Q3. What is true AND false?

Q4. What does int a=2; a=a+3; Console.WriteLine(a); output?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

sequence — Operations execute in order.

selection — Execution chooses a branch according to a condition.

iteration — A block repeats while a condition permits.

scope — Where a name can be accessed.

shadowing — A local declaration hides another accessible name.

New learning

Three control structures — Sequence runs in order; selection chooses a branch; iteration repeats instructions.

Where is a variable visible? — A local declaration belongs to its block or method; a class-level field can be accessed by relevant class methods.

Shadowing — A local variable with the same name as a field can hide that field inside the method; trace precisely which one is printed.

Worked example

A class field x is 8. A method declares int x=2 and prints x. What appears?

1. Predict first — Which x is in scope at the print statement?

2. Model solution — The method-local x is 2 and shadows the field. The program prints 2.

We Do · Practise with guidance

A class field points is 20. A method runs points += 5 and prints points without declaring a local variable.

Predict first

Which variable is modified?

Model solution

The class field is modified to 25, then 25 is printed.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Class field level = 9; method declares int level = 4 and prints level. What appears?

Question 2

Class field attempts = 2; method executes attempts++; then prints attempts. What appears?

Question 3

Describe one risk of unnecessary global/class-level mutable variables.

Exit Ticket · Answer before checking

A local variable always changes a same-named class field.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. O(n).

Q2. It must exclude mid and shrink the search interval.

Q3. false.

Q4. 5.

You Do pages · Independent questions

Q1. 4, because the local level shadows the field.

Q2. 3, because no local attempts was declared, so the class field is incremented.

Q3. Different methods can alter shared state unexpectedly, making dependencies, debugging and testing harder; prefer a local value or parameter where appropriate.

Exit Ticket · Model answer

False. A local of the same name usually shadows the field, leaving the field unchanged.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: